

My Notes for Python Libraries

Ioanna Stamou

Contents

1	Introduction	7
1.1	Overview of Libraries	7
2	Pandas	9
2.1	Importing Pandas	9
2.2	Core Data Structures	9
2.2.1	Series	9
2.2.2	DataFrame	9
2.3	Basic Operations	9
2.3.1	Insert and Drop Columns	9
2.4	Reading and Writing CSV Files	10
2.5	Indexing and Selection	10
2.5.1	Using <code>iloc</code>	10
2.6	Basic Statistics	10
2.7	Unique Values	10
3	Pathlib	11
3.1	Importing Pathlib	11
3.2	Creating Paths	11
3.3	Joining Paths	11
3.4	Checking Files and Directories	11
3.5	Creating Directories	12
3.6	Listing Files with <code>glob</code>	12
4	Xarray	13
4.1	Importing Xarray	13
4.2	Opening Datasets	13
4.3	Dataset Structure	13
4.4	Accessing Variables	13
4.5	Converting to NumPy Arrays	14
4.6	Selecting Data with <code>sel</code>	14
4.7	Index-Based Selection with <code>isel</code>	14
4.8	Saving Datasets	14
4.9	Key Idea	14
5	NumPy	15
5.1	Importing NumPy	15
5.2	NumPy Arrays	15
5.2.1	Array Dimensions	15
5.3	Creating Arrays	16
5.4	Indexing and Slicing	16
5.5	Array Operations	16

5.5.1	Element-wise Operations	16
5.5.2	Array-to-Array Operations	16
5.6	Mathematical Functions	17
5.7	Boolean Indexing	17
5.8	Linear Algebra (Very Common)	17
5.9	Random Numbers	17
5.10	Useful Array Transformations	18
5.11	Broadcasting (Super Important Concept)	18
5.12	Fitting with NumPy (<code>polyfit</code>)	18
6	SciPy (Numerical Methods Beyond NumPy)	19
6.1	Importing SciPy	19
6.2	Solving Integrals	19
6.2.1	Single Integral (<code>quad</code>)	19
6.2.2	Integrating Discrete Data (<code>trapz</code>)	20
6.3	Solving ODEs (Differential Equations)	20
6.4	Curve Fitting (General Fit)	20
6.5	Optimization (Finding Min/Max)	21
7	Matplotlib	23
7.1	Importing Matplotlib	23
7.2	Basic Plot	23
7.3	Multiple Lines on the Same Plot	23
7.4	Line Styles and Colors	24
7.5	Scatter Plots	24
7.6	Bar Plots	24
7.7	Histograms	24
7.8	Subplots	25
7.9	Figure Size and Resolution	25
7.10	Saving Figures	25
7.11	Why Matplotlib Is Important	26
8	Requests and FastAPI	27
8.1	Requests	27
8.1.1	Importing Requests	27
8.1.2	GET Request	27
8.1.3	POST Request	27
8.2	FastAPI	28
8.2.1	Basic API	28
8.2.2	Path Parameters	28
9	PyTest	29
9.1	Basic Test	29
9.2	Running Tests	29
9.3	Testing Exceptions	29
10	Scikit-learn (Machine Learning Routines)	31
10.1	Importing Scikit-learn	31
10.2	Typical Machine Learning Workflow	31
10.3	Train-Test Split	31
10.4	Data Preprocessing	32
10.4.1	Feature Scaling	32

10.4.2	Encoding Categorical Variables	32
10.5	Supervised Learning	32
10.5.1	Linear Regression	32
10.5.2	Logistic Regression	33
10.5.3	K-Nearest Neighbors (KNN)	33
10.5.4	Decision Trees	33
10.5.5	Random Forests	33
10.6	Unsupervised Learning	33
10.6.1	K-Means Clustering	33
10.6.2	Principal Component Analysis (PCA)	34
10.7	Model Evaluation	34
10.7.1	Accuracy	34
10.7.2	Confusion Matrix	34
10.7.3	Classification Report	34
10.8	Pipelines	34
10.9	Cross-Validation	35
10.10	Why Scikit-learn Is Important	35
11	PyTorch (Deep Learning Framework)	37
11.1	Installing PyTorch	37
11.2	Importing PyTorch	37
11.3	Tensors	38
11.4	Tensor Shape and Properties	38
11.5	Tensor Operations	38
11.6	Matrix Multiplication	39
11.7	Automatic Differentiation (Autograd)	39
11.8	Neural Network Modules	39
11.9	Building a Neural Network	40
11.10	Forward Pass	40
11.11	Loss Functions	40
11.12	Optimizers	40
11.13	Training Loop	41
11.14	Embedding Layers	41
11.15	Example: Tiny Language Model	41
11.16	Generating Predictions	42
11.17	Saving and Loading Models	42
11.18	Why PyTorch Is Important	42
12	PySpark	43
12.1	What Is Spark?	43
12.2	Starting a Spark Session	43
12.3	Creating DataFrames	43
12.4	Reading Data from Files	44
12.5	Basic DataFrame Operations	44
12.6	Spark Transformations vs Actions	44
12.7	Spark SQL	44
12.8	Why Use PySpark?	45

13 System and Communication Libraries	47
13.1 socket	47
13.2 pyserial	47
13.3 InfluxDB	47
13.4 Paramiko	48
13.5 subprocess	48

Chapter 1

Introduction

This document contains short notes about the Python libraries. Each chapter introduces a library, explains what it is mainly used for, and provides small examples.

1.1 Overview of Libraries

- **Pandas:** data processing and data analysis with tables (DataFrames)
- **Pathlib:** python library for handling path
- **xarray:** data processing library for multi-dimensional labeled data
- **NumPy:** numerical computing, arrays, and mathematical operations
- **SciPy:** scientific computing (ODE solving, integration, optimization, curve fitting)
- **Matplotlib:** visualization and plotting (line plots, histograms, scatter plots, etc.)
- **Requests:** sending HTTP requests to APIs (GET, POST, etc.)
- **FastAPI:** building backend APIs (server-side endpoints)
- **PyTest:** testing Python code (unit tests, exception tests)
- **Scikit-learn:** machine learning routines (preprocessing, training, evaluation)
- **PyTorch:** deep learning framework for neural networks, automatic differentiation, and GPU-accelerated tensor computation
- **PySpark:** big data processing using Apache Spark (distributed DataFrames and SQL)
- **socket:** low-level network communication
- **pyserial:** serial communication with hardware devices
- **InfluxDB client:** writing and querying time-series data for monitoring/metrics
- **Paramiko:** SSH connections and remote command execution
- **subprocess:** running external system commands from Python

These notes are meant to be a practical reference: short explanations, clean examples, and the most common routines.

Chapter 2

Pandas

Pandas is a Python library for **data processing and data analysis**. It provides high-level data structures that make working with tabular data easy and intuitive.

2.1 Importing Pandas

```
import pandas as pd
```

The alias `pd` is a community standard and is used in almost all pandas code.

2.2 Core Data Structures

2.2.1 Series

A **Series** is a one-dimensional labeled array. It is similar to a column in a table.

```
s = pd.Series([10, 20, 30])
```

2.2.2 DataFrame

A **DataFrame** is a two-dimensional table made of rows and columns. It is the most commonly used pandas structure.

```
df = pd.DataFrame({
    "A": [1, 2, 3],
    "B": [4, 5, 6]
})
```

2.3 Basic Operations

2.3.1 Insert and Drop Columns

```
df["C"] = [7, 8, 9]      # insert column
df = df.drop("B", axis=1) # drop column
```

`axis=1` refers to columns, while `axis=0` refers to rows.

2.4 Reading and Writing CSV Files

```
df = pd.read_csv("data.csv")
df.to_csv("output.csv", index=False)
```

CSV files are one of the most common formats for tabular data.

2.5 Indexing and Selection

2.5.1 Using `iloc`

`iloc` is used for **integer-based** indexing.

```
df.iloc[0]           # first row
df.iloc[0, 1]        # row 0, column 1
df.iloc[:, 0]        # all rows, first column
```

2.6 Basic Statistics

```
df.mean()
df.mode()
df.sum()
df.max()
df.min()
```

These functions operate column-wise by default.

2.7 Unique Values

```
df["A"].unique()
df["A"].nunique()
```

`unique()` returns the values, while `nunique()` returns how many unique values exist.

Chapter 3

Pathlib

Pathlib is a Python library used for **handling file paths in a clean and safe way**. It replaces traditional string-based paths with object-oriented paths.

3.1 Importing Pathlib

```
from pathlib import Path
```

`Path` is the main class used to create and manipulate file and directory paths.

3.2 Creating Paths

```
p = Path("data/file.txt")
```

A **Path object** represents a file or directory location.

3.3 Joining Paths

```
folder = Path("data")  
file = folder / "raw" / "file.nc"
```

The `/` operator is used to join paths in a clean and readable way.

3.4 Checking Files and Directories

```
file.exists()  
file.is_file()  
file.is_dir()
```

These methods allow checking whether a path exists and what type it is.

3.5 Creating Directories

```
Path("data/processed").mkdir(parents=True, exist_ok=True)
```

`parents=True` creates all missing parent folders. `exist_ok=True` avoids errors if the folder already exists.

3.6 Listing Files with glob

```
folder = Path("data/raw")  
files = folder.glob("*.nc")
```

`glob()` searches for files that match a pattern. For example, `*.nc` means all NetCDF files.

```
for f in files:  
    print(f.name)
```

Each element returned by `glob()` is a `Path` object.

Chapter 4

Xarray

Xarray is a Python library for working with **multi-dimensional labeled data**. It is especially useful for scientific data such as weather, climate, and satellite data.

4.1 Importing Xarray

```
import xarray as xr
```

The alias `xr` is commonly used for `xarray`.

4.2 Opening Datasets

```
ds = xr.open_dataset("file.nc")
```

This loads a dataset from a NetCDF or GRIB file into an `xarray` object.

4.3 Dataset Structure

An `xarray` Dataset contains:

- **Dimensions:** axes like latitude, longitude, time
- **Coordinates:** labeled values for dimensions
- **Data variables:** actual data (e.g., `u10`, `v10`)

```
print(ds)
```

4.4 Accessing Variables

```
ds["u10"]  
ds["v10"]
```

Variables are accessed like dictionary keys.

4.5 Converting to NumPy Arrays

```
u10 = ds["u10"].values
```

`.values` extracts the raw NumPy array from the xarray object.

4.6 Selecting Data with `sel`

```
subset = ds.sel(  
    longitude=slice(2, 7),  
    latitude=slice(52, 49)  
)
```

`sel()` selects data using coordinate values (labels), not indices.

4.7 Index-Based Selection with `isel`

```
subset = ds.isel(latitude=0)
```

`isel()` selects data using integer indices.

4.8 Saving Datasets

```
ds.to_netcdf("output.nc")
```

This saves the dataset to a NetCDF file.

4.9 Key Idea

Xarray combines the power of NumPy arrays with labeled dimensions, making it ideal for structured scientific data.

Chapter 5

NumPy

NumPy (Numerical Python) is a core Python library for **numerical computing**. It provides:

- Fast numerical operations
- Multi-dimensional arrays
- Mathematical and statistical functions

NumPy is the foundation on which pandas and many scientific libraries are built.

5.1 Importing NumPy

```
import numpy as np
```

The alias `np` is the standard convention used in almost all NumPy code.

5.2 NumPy Arrays

A **NumPy array** is a fixed-size, homogeneous collection of numbers. Unlike Python lists, all elements have the same data type.

```
a = np.array([1, 2, 3, 4])
```

5.2.1 Array Dimensions

```
a.ndim      # number of dimensions
a.shape     # size along each dimension
a.size      # total number of elements
```

Arrays can be 1D (vectors), 2D (matrices), or higher-dimensional.

5.3 Creating Arrays

```
np.zeros(5)
np.ones((2, 3))
np.arange(0, 10, 2)
np.linspace(0, 1, 5)
```

- `zeros`, `ones`: initialize arrays
- `arange`: integer ranges
- `linspace`: evenly spaced values

5.4 Indexing and Slicing

```
a[0]      # first element
a[1:3]    # slice
a[-1]     # last element
```

```
b = np.array([[1, 2, 3],
              [4, 5, 6]])

b[0, 1]    # row 0, column 1
b[:, 0]    # all rows, first column
```

Indexing in NumPy is zero-based and very similar to pandas `iloc`.

5.5 Array Operations

5.5.1 Element-wise Operations

```
a + 2
a * 3
a ** 2
```

Operations are applied **element-wise**, without explicit loops.

5.5.2 Array-to-Array Operations

```
x = np.array([1, 2, 3])
y = np.array([4, 5, 6])

x + y
x * y
```


5.6 Mathematical Functions

```
np.mean(a)
np.sum(a)
np.max(a)
np.min(a)
np.std(a)
```

These functions are optimized and much faster than manual Python loops.

5.7 Boolean Indexing

```
a[a > 2]
```

Boolean indexing allows filtering arrays using conditions. This concept appears frequently in pandas as well.

5.8 Linear Algebra (Very Common)

NumPy provides many useful linear algebra tools (vectors, matrices, systems of equations). They are used everywhere in data science, physics, and machine learning.

```
A = np.array([[2, 1],
              [1, 3]])
b = np.array([1, 2])

np.dot(A, b)           # matrix-vector product
np.linalg.det(A)       # determinant
np.linalg.inv(A)       # inverse (if it exists)
x = np.linalg.solve(A, b) # solve A x = b
```

`np.linalg.solve(A, b)` is preferred over computing the inverse explicitly. It is more stable and usually faster.

5.9 Random Numbers

Random numbers are used in simulation, statistics, and machine learning. NumPy provides a modern random generator through `np.random`.

```
rng = np.random.default_rng(seed=42)

rng.random(5)           # 5 random floats in [0, 1)
rng.integers(0, 10, size=5) # random integers
rng.normal(loc=0, scale=1, size=5) # normal distribution
```

Using `default_rng` is the recommended modern approach.

5.10 Useful Array Transformations

```
a = np.array([1, 2, 3, 4, 5, 6])

a.reshape((2, 3))      # change shape
a.flatten()            # make 1D copy
a.T                   # transpose (useful for 2D arrays)
```

`reshape` is extremely common when preparing data for ML models.

5.11 Broadcasting (Super Important Concept)

Broadcasting is NumPy's rule system that lets arrays of different shapes work together in operations (without manual loops).

```
a = np.array([1, 2, 3])
b = np.array([[10],
              [20]])

a + b
```

Here, `a` is automatically "stretched" across rows to match `b`. This is one of the reasons NumPy code can be very short and fast.

5.12 Fitting with NumPy (`polyfit`)

NumPy can perform simple polynomial fitting using `np.polyfit`. For general curve fitting (any function), SciPy is typically used.

```
x = np.array([0, 1, 2, 3])
y = np.array([1, 2, 1, 3])

coeffs = np.polyfit(x, y, deg=2) # quadratic fit
p = np.poly1d(coeffs)

p(1.5) # evaluate fitted polynomial at x=1.5
```

`coeffs` contains the polynomial coefficients. `poly1d` creates a polynomial function object.

Chapter 6

SciPy (Numerical Methods Beyond NumPy)

SciPy is built on top of NumPy. It provides advanced scientific computing tools such as:

- Solving ODEs (differential equations)
- Numerical integration
- Optimization and curve fitting

6.1 Importing SciPy

```
from scipy import integrate
from scipy import optimize
```

SciPy is organized into submodules. For example, `integrate` for ODEs/integrals and `optimize` for fitting.

6.2 Solving Integrals

6.2.1 Single Integral (quad)

`integrate.quad` numerically computes:

$$\int_a^b f(x) dx$$

```
import numpy as np
from scipy import integrate

f = lambda x: np.sin(x)

result, error = integrate.quad(f, 0, np.pi)
result
```

`quad` returns the integral result and an estimate of numerical error.

6.2.2 Integrating Discrete Data (`trapz`)

```
x = np.linspace(0, np.pi, 100)
y = np.sin(x)

np.trapz(y, x)    # trapezoidal rule
```

`np.trapz` is useful when you already have sampled data points.

6.3 Solving ODEs (Differential Equations)

ODE solving is usually done with `integrate.solve_ivp`. It solves problems like:

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

```
import numpy as np
from scipy.integrate import solve_ivp

def dydt(t, y):
    return -2 * y    # dy/dt = -2y

t_span = (0, 5)
y0 = [1]

sol = solve_ivp(dydt, t_span, y0, t_eval=np.linspace(0, 5, 50))

sol.t    # time points
sol.y[0] # solution y(t)
```

`solve_ivp` is very flexible and can handle many ODE systems. The output `sol.y` is shaped as `(num_equations, num_time_points)`.

6.4 Curve Fitting (General Fit)

For fitting a custom function to data, SciPy provides `optimize.curve_fit`. It estimates parameters by minimizing least-squares error.

```
import numpy as np
from scipy.optimize import curve_fit

def model(x, a, b):
    return a * np.exp(b * x)

x = np.array([0, 1, 2, 3])
y = np.array([1.0, 2.7, 7.4, 20.1])
```

```
params, covariance = curve_fit(model, x, y)

params # [a, b]
```

`params` contains the best-fit parameters. `covariance` gives uncertainty information about the fitted parameters.

6.5 Optimization (Finding Min/Max)

Optimization means finding a value of x that minimizes or maximizes a function. SciPy provides many tools, including `minimize`.

```
import numpy as np
from scipy.optimize import minimize

f = lambda x: (x - 3)**2 + 1

result = minimize(f, x0=0)

result.x # location of minimum
result.fun # minimum value
```

`x0` is the initial guess. Some optimization algorithms need a good starting point.

Chapter 7

Matplotlib

Matplotlib is a Python library used for creating static, animated, and interactive visualizations.
It is the most fundamental plotting library in Python.

7.1 Importing Matplotlib

```
import matplotlib.pyplot as plt
```

7.2 Basic Plot

```
x = [1, 2, 3]
y = [4, 5, 6]

plt.plot(x, y)
plt.xlabel("X axis")
plt.ylabel("Y axis")
plt.title("Simple Line Plot")
plt.show()
```

`plt.show()` renders the plot. Without it, the figure may not appear.

7.3 Multiple Lines on the Same Plot

Matplotlib allows plotting multiple datasets on the same figure. Each call to `plt.plot()` adds a new line.

```
x = [1, 2, 3]
y1 = [1, 4, 9]
y2 = [1, 2, 3]

plt.plot(x, y1, label="x squared")
plt.plot(x, y2, label="x linear")
plt.legend()
plt.show()
```

`plt.legend()` displays labels defined using the `label` argument.

7.4 Line Styles and Colors

```
plt.plot(x, y1, color="blue", linestyle="--", marker="o")
plt.plot(x, y2, color="red", linestyle="-")
plt.show()
```

Common style options:

- `color`: line color
- `linestyle`: solid, dashed, dotted
- `marker`: points on the line

7.5 Scatter Plots

Scatter plots are used to visualize relationships between variables. They are very common in data analysis.

```
plt.scatter(x, y1)
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Scatter Plot Example")
plt.show()
```

7.6 Bar Plots

Bar plots are useful for comparing discrete categories.

```
categories = ["A", "B", "C"]
values = [10, 15, 7]

plt.bar(categories, values)
plt.ylabel("Value")
plt.title("Bar Plot")
plt.show()
```

7.7 Histograms

Histograms show the distribution of numerical data. They are often used in statistics and exploratory data analysis.


```
import numpy as np

data = np.random.normal(0, 1, 1000)

plt.hist(data, bins=30)
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.title("Histogram")
plt.show()
```

The `bins` parameter controls how many bars the histogram has.

7.8 Subplots

Subplots allow multiple plots in the same figure. They are useful for comparing results side by side.

```
fig, axes = plt.subplots(1, 2)

axes[0].plot(x, y1)
axes[0].set_title("Plot 1")

axes[1].plot(x, y2)
axes[1].set_title("Plot 2")

plt.show()
```

`axes` is an array of plotting areas. Each axis behaves like its own mini-plot.

7.9 Figure Size and Resolution

```
plt.figure(figsize=(6, 4), dpi=100)
plt.plot(x, y1)
plt.show()
```

`figsize` controls the size in inches. `dpi` controls resolution (dots per inch).

7.10 Saving Figures

Plots can be saved to files instead of (or in addition to) being shown. This is essential for reports and publications.

```
plt.plot(x, y1)
plt.savefig("my_plot.png")
plt.show()
```

Always call `savefig` before `show` to avoid empty images.

7.11 Why Matplotlib Is Important

- It is the foundation of most Python visualization libraries
- Almost all scientific Python code uses it
- Understanding Matplotlib makes Seaborn easier
- It is widely accepted in academic and professional work

Chapter 8

Requests and FastAPI

Modern Python applications often communicate over the network. Two key libraries are:

- **requests**: for sending HTTP requests (client-side)
- **FastAPI**: for building APIs (server-side)

8.1 Requests

requests is a simple and powerful library for making HTTP requests (GET, POST, PUT, DELETE).

8.1.1 Importing Requests

```
import requests
```

8.1.2 GET Request

```
response = requests.get("https://api.example.com/data")

response.status_code
response.json()
```

`response.json()` parses JSON responses into Python dictionaries.

8.1.3 POST Request

```
data = {"name": "Alice", "age": 30}

response = requests.post(
    "https://api.example.com/users",
    json=data
)
```

8.2 FastAPI

FastAPI is a modern Python framework for building fast, type-safe web APIs. It is widely used in backend and ML services.

8.2.1 Basic API

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def read_root():
    return {"message": "Hello World"}
```

8.2.2 Path Parameters

```
@app.get("/items/{item_id}")
def read_item(item_id: int):
    return {"item_id": item_id}
```

FastAPI automatically validates types and generates documentation.

Chapter 9

PyTest

PyTest is a testing framework for Python. It allows writing simple, readable tests for functions and modules.

9.1 Basic Test

```
def add(a, b):  
    return a + b  
  
def test_add():  
    assert add(2, 3) == 5
```

9.2 Running Tests

```
pytest
```

PyTest automatically discovers files starting with `test_`.

9.3 Testing Exceptions

```
import pytest  
  
def test_error():  
    with pytest.raises(ValueError):  
        int("abc")
```


Chapter 10

Scikit-learn (Machine Learning Routines)

Scikit-learn is the most widely used Python library for **classical machine learning**. It provides:

- Data preprocessing tools
- Supervised learning algorithms
- Unsupervised learning algorithms
- Model evaluation utilities

10.1 Importing Scikit-learn

```
import numpy as np
from sklearn import datasets
```

Scikit-learn is organized into submodules such as `model_selection`, `preprocessing`, and `metrics`.

10.2 Typical Machine Learning Workflow

Most scikit-learn workflows follow these steps:

1. Load data
2. Split into train and test sets
3. Preprocess data
4. Train model
5. Evaluate model

10.3 Train-Test Split

```
from sklearn.model_selection import train_test_split

X = np.array([[1], [2], [3], [4]])
y = np.array([0, 0, 1, 1])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
```

The test set is used only for evaluation, never for training.

10.4 Data Preprocessing

10.4.1 Feature Scaling

Many ML models require features to be on similar scales.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_train)
```

`fit` learns parameters from data, `transform` applies the transformation.

10.4.2 Encoding Categorical Variables

```
from sklearn.preprocessing import OneHotEncoder

encoder = OneHotEncoder(sparse=False)
encoded = encoder.fit_transform([["red"], ["blue"], ["red"]])
```

10.5 Supervised Learning

10.5.1 Linear Regression

Linear regression is used for predicting continuous values.

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

predictions = model.predict(X_test)
```


10.5.2 Logistic Regression

Logistic regression is used for binary classification.

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

10.5.3 K-Nearest Neighbors (KNN)

```
from sklearn.neighbors import KNeighborsClassifier

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

10.5.4 Decision Trees

```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(max_depth=3)
model.fit(X_train, y_train)
```

Decision trees are intuitive but can easily overfit.

10.5.5 Random Forests

Random forests combine many decision trees to improve performance.

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(n_estimators=100)
model.fit(X_train, y_train)
```

10.6 Unsupervised Learning

10.6.1 K-Means Clustering

Clustering groups data points without labels.

```
from sklearn.cluster import KMeans

model = KMeans(n_clusters=2)
clusters = model.fit_predict(X)
```

10.6.2 Principal Component Analysis (PCA)

PCA reduces dimensionality while preserving variance.

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
X_reduced = pca.fit_transform(X)
```

10.7 Model Evaluation

10.7.1 Accuracy

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, predictions)
```

10.7.2 Confusion Matrix

```
from sklearn.metrics import confusion_matrix

confusion_matrix(y_test, predictions)
```

10.7.3 Classification Report

```
from sklearn.metrics import classification_report

print(classification_report(y_test, predictions))
```

The classification report includes precision, recall, and F1-score.

10.8 Pipelines

Pipelines chain preprocessing and modeling steps safely. They help avoid data leakage.

```
from sklearn.pipeline import Pipeline

pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("model", LogisticRegression())
])

pipe.fit(X_train, y_train)
pipe.predict(X_test)
```

10.9 Cross-Validation

Cross-validation evaluates models more reliably than a single split.

```
from sklearn.model_selection import cross_val_score

scores = cross_val_score(model, X, y, cv=5)
scores.mean()
```

10.10 Why Scikit-learn Is Important

- Industry standard for classical ML
- Simple and consistent API
- Excellent documentation
- Ideal for rapid prototyping and research

Chapter 11

PyTorch (Deep Learning Framework)

PyTorch is one of the most widely used libraries for deep learning and neural networks.

It provides:

- GPU-accelerated tensor operations
- Automatic differentiation
- Neural network modules
- Optimizers and training utilities

It is heavily used for:

- computer vision
- language models
- reinforcement learning
- scientific machine learning

11.1 Installing PyTorch

Install PyTorch using pip:

```
pip install torch
```

Check installation:

```
import torch
print(torch.__version__)
```

11.2 Importing PyTorch

```
import torch
import torch.nn as nn
import torch.nn.functional as F
```

Typical aliases:

- torch → main library
- nn → neural network modules
- F → functional utilities

11.3 Tensors

Tensors are the core data structure of PyTorch. They are similar to NumPy arrays but support GPU acceleration.

Create tensors:

```
x = torch.tensor([1,2,3])
```

Random tensors:

```
torch.rand(3)
torch.randn(3)
```

Zeros and ones:

```
torch.zeros(5)
torch.ones(5)
```

11.4 Tensor Shape and Properties

```
x.shape
x.ndim
x.dtype
```

Example:

```
x = torch.tensor([[1,2],[3,4]])

x.shape → (2,2)
x.ndim → 2
```

11.5 Tensor Operations

Element-wise operations:

```
x + 2
x * 3
x ** 2
```

Tensor operations:

```
a = torch.tensor([1,2,3])
b = torch.tensor([4,5,6])

a + b
a * b
```

11.6 Matrix Multiplication

Very common in neural networks.

```
A = torch.tensor([[1,2],[3,4]])
B = torch.tensor([[5,6],[7,8]])

torch.matmul(A,B)
```

11.7 Automatic Differentiation (Autograd)

PyTorch automatically computes gradients.

```
x = torch.tensor(3.0, requires_grad=True)

y = x**2

y.backward()

print(x.grad)
```

Output:

6

because:

$$d(x^2)/dx = 2x$$

11.8 Neural Network Modules

PyTorch provides many layers.

Common layers:

```
nn.Linear
nn.Embedding
nn.Conv2d
nn.LSTM
```

Example:

```
layer = nn.Linear(10,5)
```

This maps:

10 inputs → 5 outputs

11.9 Building a Neural Network

Neural networks are defined as classes.

```
class SimpleModel(nn.Module):  
  
    def __init__(self):  
        super().__init__()  
  
        self.linear = nn.Linear(10,1)  
  
    def forward(self, x):  
  
        output = self.linear(x)  
  
        return output
```

11.10 Forward Pass

The forward() method defines how data flows through the model.

```
model = SimpleModel()  
  
x = torch.rand(1,10)  
  
y = model(x)
```

11.11 Loss Functions

Loss measures how wrong the model prediction is.

Common losses:

```
nn.MSELoss  
nn.CrossEntropyLoss  
nn.BCELoss
```

Example:

```
loss_fn = nn.MSELoss()  
loss = loss_fn(predictions, targets)
```

11.12 Optimizers

Optimizers update model weights.

Common optimizers:

```
SGD  
Adam  
RMSprop
```

Example:

```
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
```


11.13 Training Loop

The standard PyTorch training loop:

```
for epoch in range(epochs):  
  
    optimizer.zero_grad()  
  
    predictions = model(X)  
  
    loss = loss_fn(predictions, y)  
  
    loss.backward()  
  
    optimizer.step()
```

Explanation:

```
zero_grad() → clear gradients  
forward pass → compute predictions  
loss → compute error  
backward() → compute gradients  
step() → update weights
```

11.14 Embedding Layers

Embeddings convert token IDs into vectors.

Used in language models.

```
embedding = nn.Embedding(vocab_size, embedding_dim)
```

Example:

```
token_id = 5  
→ vector representation
```

11.15 Example: Tiny Language Model

Example architecture used in the tiny language model.

```
class TinyLanguageModel(nn.Module):  
  
    def __init__(self, vocab_size, embed_dim):  
  
        super().__init__()  
  
        self.embedding = nn.Embedding(vocab_size, embed_dim)  
  
        self.linear = nn.Linear(embed_dim, vocab_size)  
  
    def forward(self, input_ids):
```

```

x = self.embedding(input_ids)

x = x.mean(dim=1)

logits = self.linear(x)

return logits

```

Pipeline:

```

tokens
↓
embedding
↓
sentence vector
↓
linear layer
↓
token probabilities

```

11.16 Generating Predictions

After training, we can generate predictions.

with `torch.no_grad()`:

```

logits = model(input_ids)

probs = torch.softmax(logits, dim=-1)

next_token = torch.argmax(probs)

```

11.17 Saving and Loading Models

Save model:

```
torch.save(model.state_dict(), "model.pt")
```

Load model:

```
model.load_state_dict(torch.load("model.pt"))
```

11.18 Why PyTorch Is Important

PyTorch is widely used because:

- simple Pythonic design
- strong research community
- flexible dynamic computation graphs
- widely used in modern AI systems

Frameworks like GPT, LLaMA, Stable Diffusion, and many research models are built with PyTorch.

Chapter 12

PySpark

PySpark is the Python interface to **Apache Spark**, a distributed computing framework used for processing large datasets. PySpark allows Python users to leverage Spark's speed and scalability.

12.1 What Is Spark?

Apache Spark is designed for:

- Distributed data processing
- In-memory computation
- Large-scale data analytics

It is commonly used when data is too large to fit in memory on a single machine.

12.2 Starting a Spark Session

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("MyApp") \
    .getOrCreate()
```

The `SparkSession` is the main entry point to PySpark.

12.3 Creating DataFrames

```
data = [(1, "Alice"), (2, "Bob"), (3, "Charlie")]
columns = ["id", "name"]

df = spark.createDataFrame(data, columns)
df.show()
```

Spark DataFrames are similar to pandas DataFrames, but they operate in a distributed way.

12.4 Reading Data from Files

```
df = spark.read.csv(  
    "data.csv",  
    header=True,  
    inferSchema=True  
)
```

Spark supports CSV, JSON, Parquet, ORC, and many other formats.

12.5 Basic DataFrame Operations

```
df.select("name").show()  
df.filter(df["id"] > 1).show()  
df.groupBy("name").count().show()
```

12.6 Spark Transformations vs Actions

Spark operations are divided into:

- **Transformations:** define what to do (lazy)
- **Actions:** trigger computation

```
df_filtered = df.filter(df["id"] > 1) # transformation  
df_filtered.show()                   # action
```

12.7 Spark SQL

```
df.createOrReplaceTempView("people")  
  
spark.sql("""  
    SELECT name, COUNT(*)  
    FROM people  
    GROUP BY name  
""").show()
```

Spark SQL allows querying DataFrames using SQL syntax.

12.8 Why Use PySpark?

- Handles very large datasets
- Scales across clusters
- Faster than traditional disk-based systems
- Integrates well with Hadoop

Chapter 13

System and Communication Libraries

These libraries allow Python to interact with the operating system, networks, hardware devices, and external services. They are commonly used in backend and infrastructure software.

13.1 socket

The `socket` module enables low-level network communication.

```
import socket

s = socket.socket()
s.connect(("localhost", 8080))
s.send(b"Hello")
s.close()
```

13.2 pyserial

`pyserial` is used for serial communication with hardware devices.

```
import serial

ser = serial.Serial("/dev/ttyUSB0", baudrate=9600)
data = ser.readline()
ser.close()
```

13.3 InfluxDB

InfluxDB is a time-series database used for monitoring and metrics. Python clients allow writing and querying time-based data.

```
from influxdb_client import InfluxDBClient

client = InfluxDBClient(url="http://localhost:8086", token="TOKEN")
```

13.4 Paramiko

paramiko is used for SSH connections and remote command execution.

```
import paramiko

ssh = paramiko.SSHClient()
ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
ssh.connect("server.com", username="user")
stdin, stdout, stderr = ssh.exec_command("ls")
ssh.close()
```

13.5 subprocess

The `subprocess` module allows running external system commands. It replaces older modules like `os.system`.

```
import subprocess

result = subprocess.run(
    ["ls", "-l"],
    capture_output=True,
    text=True
)

result.stdout
```